

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 0603

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO4: K- Nearest Neighbour Learning – Locally weighted Regression – Radial Bases Functions – Case Based Learning (**BL 5**)

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

Industry Problem Solving

Code of Conduct:

I X. Joshua – (192312188) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

X. Joshua

1. Retail Product Recommendation — KNN

Solution:

Use **K-Nearest Neighbour (KNN)** to recommend products based on customers with similar purchase histories.

Working:

1. Represent each customer using features such as products purchased, purchase frequency, amount spent, and categories.
2. Normalize the features so that one feature does not dominate.
3. Calculate similarity using **Euclidean distance** or **Cosine similarity**.
4. Find the K customers most similar to the target customer.
5. Recommend products purchased by these neighbours but not yet purchased by the target customer.

Example:

If Customer A is similar to Customers B, C, and D, and B/C/D purchased a laptop bag, the system recommends the laptop bag to A.

Adaptation:

New purchases are added to the customer profile. The similarity calculation is repeated, allowing recommendations to change as customer behaviour changes.

2. Manufacturing Machine Output — Locally Weighted Regression

Solution:

Use **Locally Weighted Regression (LWR)** to predict machine output from sensor readings such as temperature, pressure, vibration, and speed.

Working:

1. Collect historical sensor readings and machine output.
2. For a new sensor reading, calculate its distance from previous observations.
3. Give higher weights to nearby observations using a Gaussian kernel:
4. Build a local regression model using these weights.
5. Predict the machine output.

Advantage:

Unlike global linear regression, LWR focuses on observations similar to the current operating condition, improving prediction when machine behaviour changes.

Noisy/incomplete

data:

Outliers can receive low weights. Missing sensor values can be handled using imputation or interpolation before prediction.

3. Healthcare Patient Risk Classification — RBF Network

Solution:

Use a **Radial Basis Function (RBF) network** to classify patients into low, medium, and high-risk categories.

Working:

1. Use features such as age, blood pressure, glucose level, cholesterol, and medical history.
2. Normalize the features.
3. Select important features using correlation, feature importance, or domain knowledge.
4. RBF neurons calculate Gaussian activation based on the distance between the patient and the RBF centers.
5. The output layer assigns the patient to a risk category.

Large**datasets:**

Training can be improved by selecting representative centers using clustering such as K-Means.

Accuracy**improvement:**

Feature normalization, appropriate centers and widths, cross-validation, and removal of irrelevant features improve classification accuracy.

4. E-Commerce Similar Product Detection — Case-Based Learning

Solution:

Use **Case-Based Learning (CBL)** to find products similar to a new product using previous product cases.

Working:

1. Store previous products with features such as price, category, brand, rating, specifications, and sales.
2. Convert product information into feature vectors.
3. Calculate similarity using Euclidean distance or Cosine similarity.
4. Retrieve the most similar products.
5. Reuse their previous pricing information to support dynamic pricing.

Example:

If a new smartphone is similar to three previously sold smartphones, their prices and sales performance can help determine a suitable price.

Scalability:

Use indexing, clustering, or approximate nearest-neighbour search to quickly retrieve similar products from millions of cases.

5. Logistics Delivery-Time Prediction — RBF Network

Solution:

Use an **RBF network** to predict delivery time using distance, traffic, weather, location, and delivery time.

Working:

1. Collect historical delivery data.
2. Normalize input features.
3. Select RBF centers representing different traffic/location conditions.
4. Calculate Gaussian activation for each center.
5. Combine RBF outputs to predict delivery time.

RBF networks are useful because they can model **non-linear relationships**.

Example:

A 10 km delivery may take 20 minutes in low traffic but 50 minutes during heavy traffic. RBF can capture this non-linear relationship.

Reliability:

Use sufficient historical data, remove outliers, update the model regularly, and evaluate using MAE/RMSE.

6. Banking Loan Default Prediction — Logistic Regression

Solution:

Use **Logistic Regression** to predict whether a customer is likely to default on a loan.

Features:

- Income
- Credit score
- Existing loans
- Debt-to-income ratio
- Transaction history
- Repayment history

Working:

1. Clean and normalize the financial data.
2. Select important features.
3. Train Logistic Regression.

4. The model calculates:
5. If probability is above a selected threshold, classify the customer as high-risk.

Adaptation:

New repayment and transaction data can periodically be added to retrain the model.

Advantage:

The decision boundary separates likely defaulters from non-defaulters and the output probability can be used for risk assessment.

7. Smart Agriculture Crop Yield — LWR

Solution:

Use **Locally Weighted Regression** to predict crop yield using soil quality, rainfall, temperature, fertilizer, and irrigation.

Working:

1. Collect historical agricultural data.
2. Normalize the features.
3. For a new field condition, calculate distances from historical observations.
4. Assign higher weights to similar environmental conditions.
5. Fit a local regression model and predict crop yield.

Seasonal**variation:**

Separate or weighted historical observations can represent different seasons.

Noisy**data:**

Sensor errors and unusual weather observations can be reduced using preprocessing, outlier removal, and suitable kernel bandwidth.

Advantage:

Local learning can provide better predictions when crop behaviour changes under different environmental conditions.

8. Cybersecurity Intrusion Detection — KNN

Solution:

Use **KNN** to identify abnormal network behaviour.

Features:

- Packet size
- Connection duration
- Number of packets

- Protocol
- Source/destination information
- Failed login attempts

Working:

1. Convert network traffic into feature vectors.
2. Normalize the features.
3. Calculate distance between new traffic and known normal/attack patterns.
4. Find K nearest examples.
5. Classify the traffic based on the majority class.

Anomaly

detection:

If a new network pattern is very far from known normal cases, it can be flagged as suspicious.

Adaptation:

New confirmed attack patterns can be added to the training dataset, allowing KNN to recognize emerging attacks.

9. Movie Recommendation — KNN

Solution:

Use **KNN** to find users with similar viewing histories and ratings.

Working:

1. Create a user-movie rating matrix.
2. Represent every user as a vector of ratings.
3. Calculate similarity using **Cosine similarity**:
4. Find the K most similar users.
5. Recommend movies highly rated by those users but not watched by the target user.

Large-scale

data:

Use sparse matrices and approximate nearest-neighbour techniques to reduce computation.

Adaptation:

When users watch or rate new movies, their profiles are updated and recommendations change accordingly.

10. Transportation Route Optimization — RBF Network

Solution:

Use an **RBF network** to estimate travel time or fuel consumption for different routes.

Inputs:

- Distance
- Traffic level
- Road type
- Vehicle speed
- Fuel consumption
- Weather

Working:

1. Collect historical route data.
2. Train an RBF network using these variables.
3. Gaussian functions identify similar traffic and road conditions.
4. Predict travel time/fuel consumption.
5. Select the route with the best predicted cost.

Advantage:

RBF can model complex non-linear relationships, such as traffic increasing fuel consumption disproportionately.

Real-time**use:**

Current traffic data can be supplied to the model to continuously update route decisions.

11. Telecommunications Data Plan Recommendation — KNN**Solution:**

Use **KNN** to recommend suitable mobile data plans.

Features:

- Monthly data usage
- Call duration
- SMS usage
- Streaming usage
- Current plan cost

Working:

1. Represent every customer using usage features.
2. Normalize the features.
3. Find customers with similar usage patterns.

4. Examine the plans used successfully by these neighbours.
5. Recommend the most suitable plan.

Example:

If a customer uses 80 GB/month and similar customers prefer a 100 GB plan, the system can recommend that plan.

Adaptation:

The customer profile is updated monthly, allowing recommendations to change as usage patterns change.

12. Electricity Consumption Prediction — RBF Network

Solution:

Use an **RBF network** to predict electricity consumption based on time, weather, and appliance usage.

Inputs:

- Hour/day
- Temperature
- Humidity
- Number of appliances
- Previous consumption

Working:

1. Collect historical electricity data.
2. Normalize input features.
3. Select RBF centers representing different consumption patterns.
4. Calculate Gaussian activations.
5. Predict electricity consumption through the output layer.

Seasonal

data:

Different centers can represent summer, winter, weekdays, weekends, and peak hours.

Noise

handling:

Remove extreme sensor errors and use sufficient historical data to make predictions more reliable.

13. Online Payment Fraud Detection — Logistic Regression

Solution:

Use **Logistic Regression** to classify transactions as legitimate or fraudulent.

Features:

- Transaction amount
- Transaction time
- Location
- Device
- Frequency of transactions
- Previous transaction behaviour

Working:

1. Clean and normalize transaction data.
2. Select relevant features.
3. Train Logistic Regression using legitimate and fraudulent transactions.
4. Calculate fraud probability using the sigmoid function.
5. Use a suitable threshold to classify the transaction.

Decision**boundary:**

Logistic Regression learns a boundary separating fraudulent and legitimate transactions.

Adaptation:

New confirmed fraud cases can be added to the training data periodically to handle changing fraud patterns.

14. Manufacturing Defect Detection — RBF Network

Solution:

Use an **RBF network** to classify products as defective or non-defective.

Inputs:

- Temperature
- Pressure
- Vibration
- Weight
- Dimensions
- Quality measurements

Working:

1. Collect sensor readings from production.
2. Normalize the data.

3. Use K-Means to select representative RBF centers.
4. Calculate Gaussian activation for each input.
5. The output layer predicts **Defective** or **Non-Defective**.

Non-linear**relationships:**

RBF can capture complex relationships between sensor values that simple linear models may miss.

Improvement:

As more production cases become available, retraining the network can improve its ability to recognize new defect patterns.

15. Smart City Traffic Prediction — KNN

Solution:

Use **KNN** to predict traffic flow and congestion using historical sensor data.

Features:

- Vehicle count
- Average speed
- Time of day
- Weather
- Day of week
- Previous traffic level

Working:

1. Collect real-time sensor readings.
2. Normalize the features.
3. Compare the current traffic condition with historical conditions.
4. Find the K most similar traffic patterns.
5. Predict the current congestion level using their outputs.

Example:

If the current traffic pattern is similar to previous Monday 9 AM patterns, their historical congestion levels can be used for prediction.

Real-time**and****scalability:**

Use recent data, efficient nearest-neighbour search, indexing, and distributed processing to handle large volumes of sensor data.